

Beta Distribution

CA2 @ EPS Course

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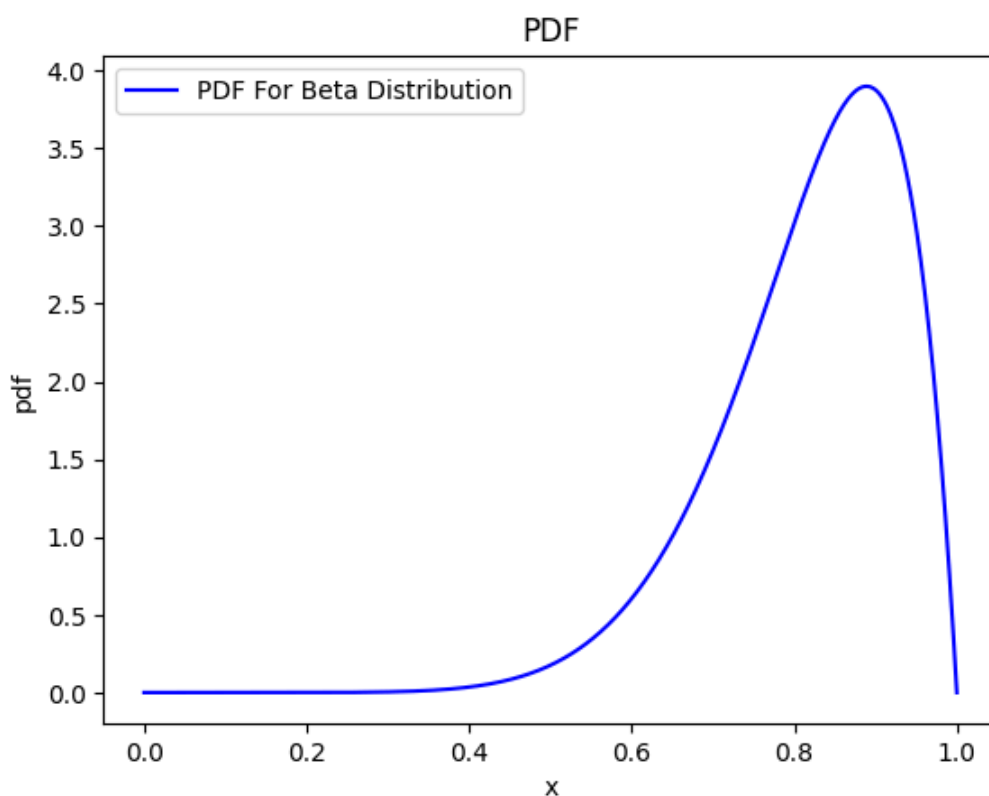
Instructions:

- Replace the placeholders (between `## TODO ##` and `## TODO ##`) with the appropriate details.
 - Ensure you run each cell after you've entered your solution.
-

Full Name: ...

SID: ...

The objective of this notebook is to estimate the probability of getting tails in a coin flip using a Bayesian approach.



Read the `coint_flips.txt` file and plot the posterior distributions after every 50 trials for two prior probabilities:

1. Uniform distribution
2. $Beta(4, 10)$

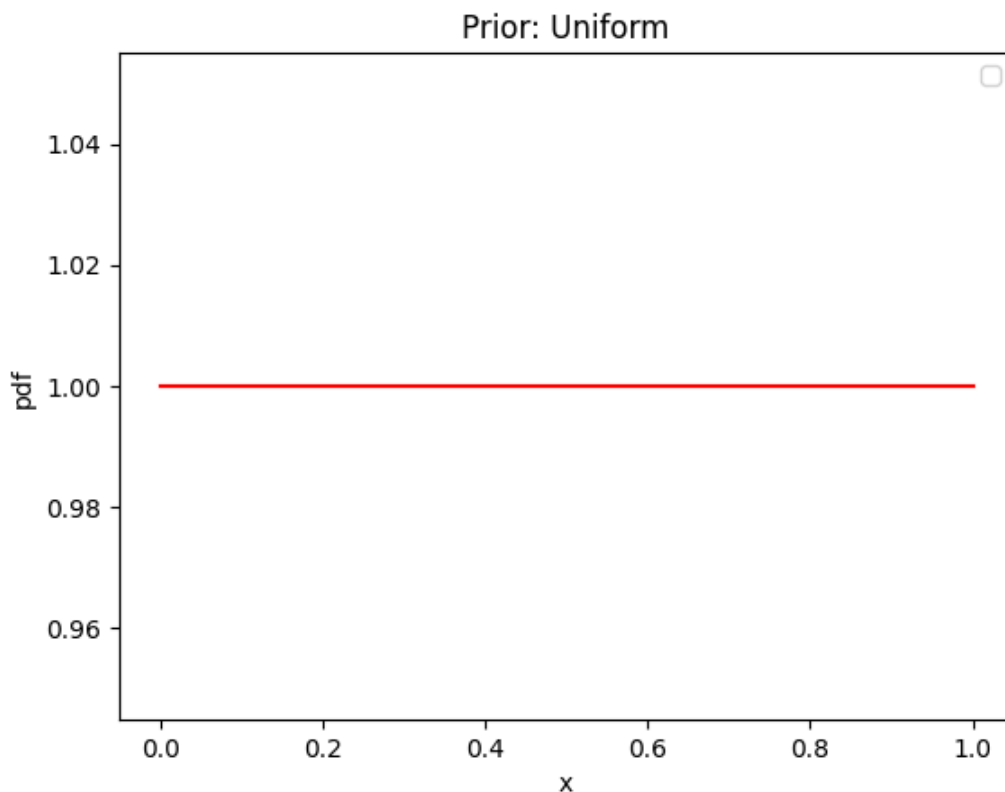
Prior: Uniform

Hint: use `plt.figure(figsize=(25, 20))` and `plt.subplot(4, 5, ...)`

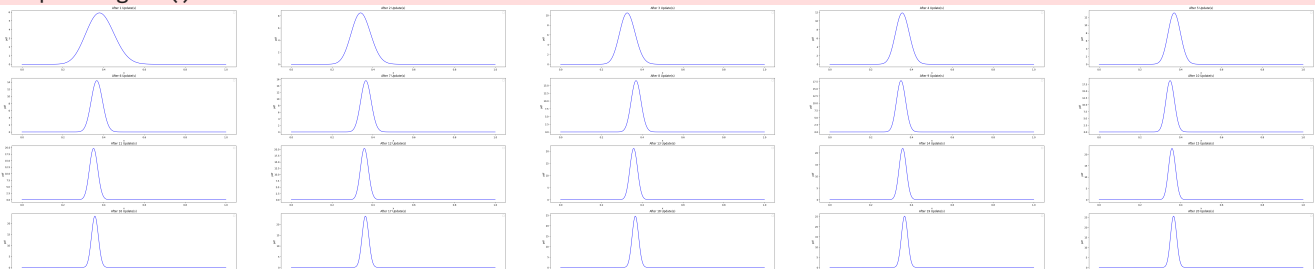
First I put a and b parameters equal to 1 because I want the initial prior distribution to be a uniform distribution ($\text{Beta}(1, 1) = U(0, 1)$). This prior distribution tells that there is no known information and we know nothing about how much the probabilities that the coin comes head or tail are in advance.

Then I plotted the prior distribution and after that in a loop I iterated over the coin flips list which I created in the former code cell. In this iteration the step is 50 actually, and I slice the list by the update rate equal to 50. Then I plot all the $1000/50 = 20$ graphs and we can see the beta distribution progressing and getting thinner and more fit to its mode by each iteration.

```
C:\Users\Shahab\AppData\Local\Temp\ipykernel_14476\3786585036.py:15: UserWarning: No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.  
plt.legend()
```



```
C:\Users\Shahab\AppData\Local\Temp\ipykernel_14476\3786585036.py:34: UserWarning: No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.  
plt.legend()
```



Now calculate the final posterior mean and variance.

Final Mean: 0.3652694611

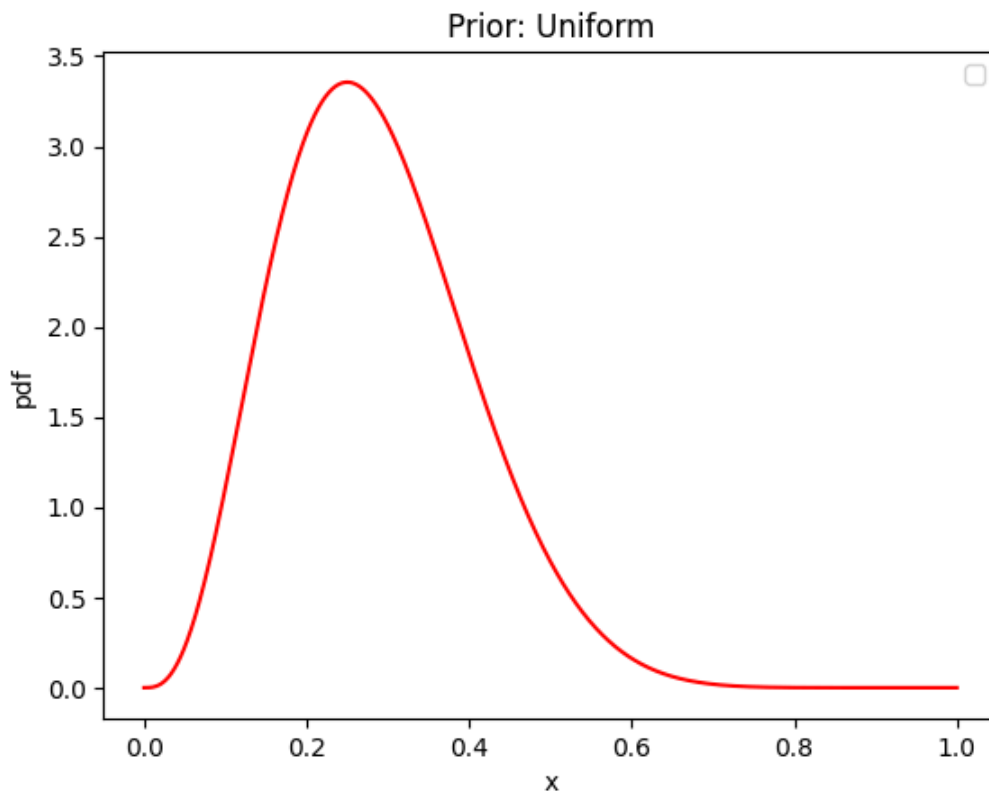
Final Variance: 0.0002311542

Prior: $Beta(4, 10)$

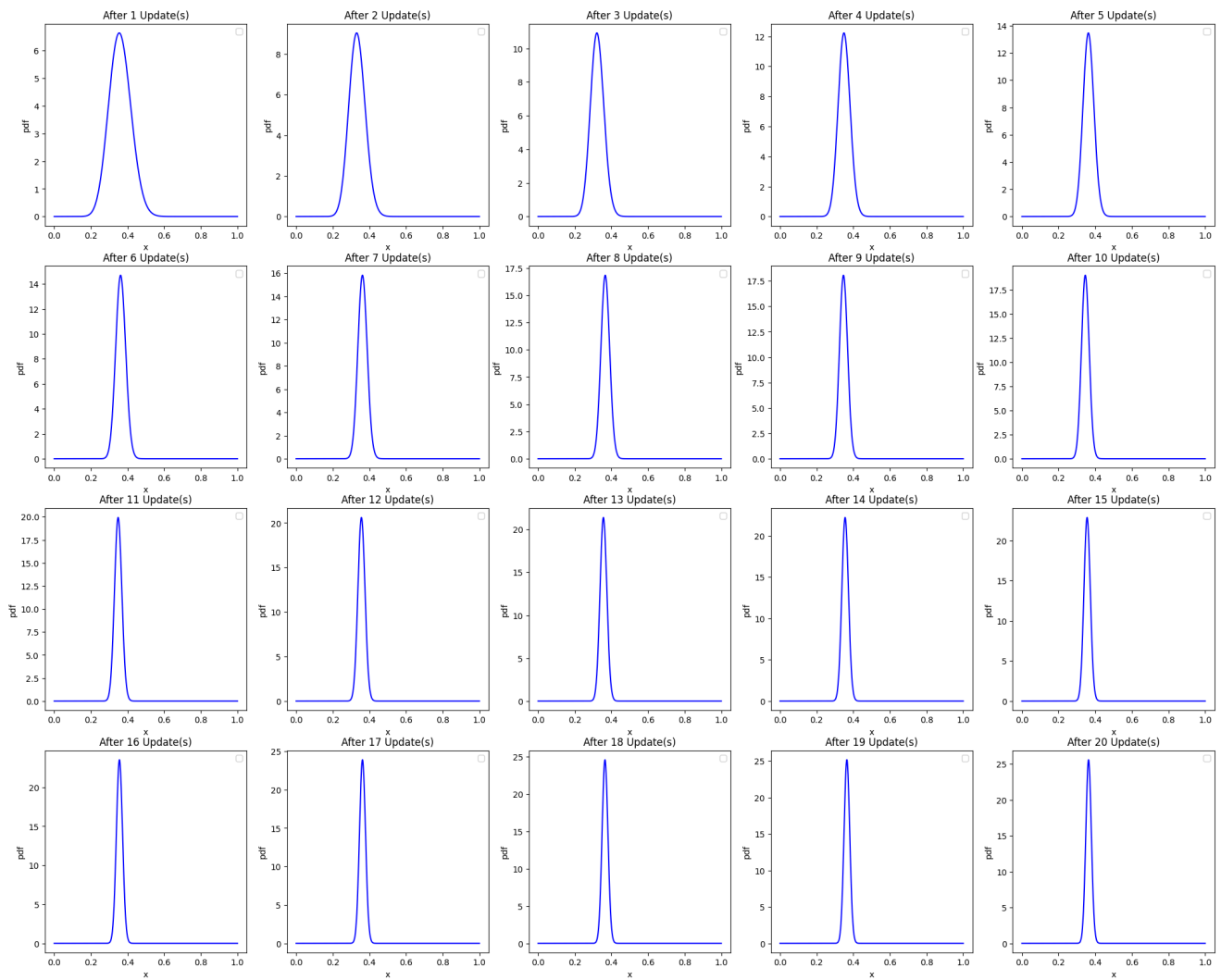
This part is exactly like the previous part, but has a slight difference. That is the prior distribution. In this part we assume that we know some information about how much the probabilities of the coin are.

This time we put the prior distribution equal to $Beta(4, 10)$.

```
C:\Users\Shahab\AppData\Local\Temp\ipykernel_14476\195806407.py:13: UserWarning: No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.  
plt.legend()
```



```
C:\Users\Shahab\AppData\Local\Temp\ipykernel_14476\195806407.py:32: UserWarning: No artists with labels found to put in legend. Note that artists whose label start with an underscore are ignored when legend() is called with no argument.  
plt.legend()
```



Now calculate the final posterior mean and variance.

Final Mean: 0.3639053254

Final Variance: 0.0002280574